Black Hole Unitarity via Reflexive Unitarity

and Stinespring Dilation:

A GKSL Model in a JT-like PSC Universe

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Abstract

We examine black hole information unitarity within the Mathematical Foundations of Reflexive Reality (MFRR) framework [14]. The MFRR Transputation operator PT^{-1} provides a physical mechanism for unitary information recovery consistent with the Perfect Self-Containment condition. Since every CPTP map admits a Stinespring dilation (a mathematical theorem independent of gravitational modelling), Hawking evaporation is unitarily equivalent to a reversible evolution on a system-plus-environment space.

We construct the evaporation channel $\Phi_{\Delta t} = e^{\Delta t \mathcal{L}}$ as a GKSL master equation for a JT-like system of $n = 3$ bosonic modes ($d_{\mathcal{H}} = 8$) at $T_H \approx 0.004$. Stinespring dilation achieves fidelity $F \geq 1 - 10^{-8}$; detailed balance holds to 0.00%; the entropy curve reaches 97.2% of the ideal thermal value. In two complementary closed unitary models — a pure initial BH state and a purified thermal initial state with reference system R — the full Page curve with turnover is reproduced and PT^{-1} recovery achieves $F = 1.000000$.

Open fronts are stated explicitly: extension to $3 + 1$ quantum gravity, the firewall tension, and a first-principles derivation of PT^{-1}.

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1 Introduction

The black hole information paradox [4, 5] stands as one of the deepest unresolved tensions in theoretical physics. Hawking's semiclassical calculation [6] predicts that an evaporating black hole emits thermal radiation, apparently destroying all correlations with the infalling matter. If correct, this would violate unitarity—the cornerstone of quantum mechanics. Page argued [11] that if information is ultimately preserved, the entanglement entropy of Hawking radiation must rise and then fall along what is now called the Page curve.

Modern approaches to this tension include: the island formula and quantum extremal surfaces [1, 12] (which reproduce the Page curve in holographic settings); the ER=EPR proposal [9], connecting horizon entanglement to non-traversable wormholes; and the firewall paradox [2], sharpening the tension between unitarity and the equivalence principle. Despite significant progress, a fully non-perturbative, background-independent resolution remains elusive.

In this paper we examine the information paradox from the perspective of the Mathematical Foundations of Reflexive Reality (MFRR) [14, 17]. The MFRR framework introduces a Perfect Self-Containment (PSC) condition and an internal adjudication operator PT governing the coherence of self-referential structures. The canonical inverse PT^{-1} provides a physical mechanism by which information encoded in a PSC universe is not merely mathematically recoverable but is structurally required to be recovered—we call this property *reflexive unitarity*.

Our central observation is that reflexive unitarity, combined with the Stinespring dilation theorem [19], yields a concrete programme: model Hawking evaporation as a GKSL (Gorini-Kossakowski- Sudarshan-Lindblad) master equation [3, 8], note that the resulting channel is CPTP and therefore admits a Stinespring dilation by theorem, and then use PT^{-1} to interpret which unitary extension is physically realised. We implement this programme explicitly in a JT-like [7, 20] PSC toy universe and report the computational results of experiment TE2.4 [13].

We emphasise from the outset what this work does and does not claim. The Stinespring theorem guarantees a unitary dilation for *any* CPTP map; this is not new. The novelties are: (i) the MFRR interpretation of PT^{-1} as the physical recovery operator, and (ii) the explicit construction and numerical verification of the Stinespring unitary for the JT-like model. Extension to realistic 3+1 quantum gravity is an open problem.

The paper is organised as follows. Section 2 introduces the PSC framework and the $\mathrm{PT}/\mathrm{PT}^{-1}$ mechanism. Section 3 reviews GKSL dynamics and Stinespring dilation and describes the JT-like model. Section 4 presents the TE2.4 results. Section 5 situates the MFRR approach relative to existing approaches. Section 6 discusses scope limitations and open fronts. Section 7 concludes. Appendices collect model parameters, construction details, and reproducibility information.

2 The PSC Framework and Reflexive Unitarity

2.1 The Perfect Self-Containment Condition

The MFRR framework [14] posits that a physically realised universe must satisfy a Perfect Self-Containment (PSC) condition: the universe's global state Ω must remain coherent under its own adjudication at every moment. Formally, the PSC condition requires that the informational structure of Ω be self-sustaining—no external selector is permitted; the universe must select its own internal laws [14].

Within this framework, Transputation (PT) is an internal adjudication operator that evaluates whether a proposed state transition is self-consistent. It acts as a kind of internal filter: only transitions that preserve PSC are realised. The operator PT is not a Hamiltonian or a density matrix; it is a higher-level structural constraint that acts on the space of possible universe histories.

2.2 The Canonical Reverse PT^{-1}

A key feature of the MFRR construction is that PT admits a canonical inverse PT^{-1}. This inverse is not merely a formal algebraic object; it represents the physical process by which a PSC universe *undoes* the apparent information loss introduced by open-system dynamics. Concretely, PT^{-1} implements the recovery channel that restores the full quantum information of the infalling matter from the Hawking radiation, *given access to the complete environment.*

The critical interpretive point is this: PT^{-1} does not violate the second law or the semiclassical approximation. Rather, it operates at the level of the full PSC universe (system plus environment), where unitary evolution is restored by the Stinespring construction. The apparent non-unitarity of the Hawking channel is, in this picture, a consequence of tracing out the environment; the underlying evolution on $\mathcal{H}_{\mathrm{sys}} \otimes \mathcal{H}_E$ remains unitary.

2.3 Reflexive Unitarity

We define *reflexive unitarity* as the property that a PSC universe, evolving under GKSL dynamics on its system Hilbert space, admits a unitary dilation on the enlarged space $\mathcal{H}_{\mathrm{sys}} \otimes \mathcal{H}_E$ such that the PT^{-1} operator coincides with the environment-tracing adjoint of this dilation. This is not an additional postulate: it follows from the Stinespring theorem (the dilation exists for any CPTP channel) together with the PSC requirement (which selects a specific physical interpretation for the environment degrees of freedom).

Definition 2.1 (Reflexive Unitarity)**.** *A quantum channel $\Phi : \mathcal{B}(\mathcal{H}_{\mathrm{sys}}) \to \mathcal{B}(\mathcal{H}_{\mathrm{sys}})$ satisfies* reflexive unitarity *with respect to a PSC adjudication pair $(\mathrm{PT}, \mathrm{PT}^{-1})$ if there exists a Hilbert space $\mathcal{H}_E$ and a unitary $U \in \mathcal{B}(\mathcal{H}_{\mathrm{sys}} \otimes \mathcal{H}_E)$ such that*

$$\Phi(\rho) = \mathrm{Tr}_E\Big[U(\rho \otimes |0\rangle\langle 0|_E)\, U^\dagger\Big]$$

and PT^{-1} is identified with the adjoint recovery map $\rho_E \mapsto \mathrm{Tr}_{\mathrm{sys}}[U^\dagger(\cdot)U]$ restricted to the environment.

Remark 2.2. The existence of U is guaranteed by the Stinespring theorem for any CPTP channel Φ [10,19]. The content of reflexive unitarity is not the existence of U but the physical identification of U with the PT^{-1} recovery process within the PSC framework. This identification is *interpretive rather than structural*: the Stinespring theorem guarantees the existence of some U for any CPTP map; PSC provides the physical rationale for which U is physically realized, without imposing additional mathematical constraints on U beyond those of the Stinespring construction. The novel content is the identification, not the existence of the dilation.

3 GKSL Dynamics and Stinespring Dilation

3.1 The Stinespring Dilation Theorem

We recall the relevant mathematical fact.

Theorem 3.1 (Stinespring 1955 [19])**.** *Let $\Phi : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ be a completely positive trace-preserving map. Then there exists a Hilbert space $\mathcal{H}_E$ and an isometry $V : \mathcal{H} \to \mathcal{H} \otimes \mathcal{H}_E$ such that*

$$\Phi(\rho) = \mathrm{Tr}_E\Big[V \rho V^\dagger\Big].$$

The isometry V extends to a unitary U on $\mathcal{H} \otimes \mathcal{H}_E$ by choosing any completion of V to a square unitary, giving $\Phi(\rho) = \mathrm{Tr}_E[U(\rho \otimes |0\rangle\langle 0|_E)U^\dagger]$.

A consequence familiar from quantum information theory [10] is the Kraus representation: for any CPTP map there exist Kraus operators $\{K_k\}$ with $\sum_k K_k^\dagger K_k = I$ such that $\Phi(\rho) = \sum_k K_k \rho K_k^\dagger$. The isometry is then $V|\psi\rangle = \sum_k K_k|\psi\rangle \otimes |k\rangle_E$, and $\dim \mathcal{H}_E$ equals the number of Kraus operators.

3.2 GKSL Master Equation

The GKSL (Lindblad) master equation [3,8] describes the most general Markovian, completely positive, trace-preserving evolution of an open quantum system:

$$\begin{aligned} \frac{d\rho}{dt} &= \mathcal{L}[\rho], \\ \mathcal{L}[\rho] &:= -i[H,\rho] + \sum_k \gamma_k \Big(L_k \rho L_k^\dagger \\ &\qquad -\tfrac{1}{2} L_k^\dagger L_k \rho - \tfrac{1}{2} \rho L_k^\dagger L_k \Big), \end{aligned} \tag{1}$$

where H is the system Hamiltonian, L_k are Lindblad jump operators, and $\gamma_k \geq 0$ are rates. The propagator $\Phi_t = e^{t\mathcal{L}}$ is a one-parameter semigroup of CPTP maps for all $t \geq 0$.

Because Φ_t is CPTP, the Stinespring theorem guarantees a unitary dilation for each t. For a small time

step Δt, the Kraus operators take the explicit form

$$K_0 = I - \left(iH + \tfrac{1}{2} \sum_k \gamma_k L_k^\dagger L_k \right) \Delta t, \tag{2}$$

$$K_k = \sqrt{\gamma_k \Delta t}\, L_k, \quad k = 1, \ldots, m. \tag{3}$$

The environment dimension for this step is $\dim \mathcal{H}_E = 1 + m$, where m is the number of distinct Lindblad operators.

Remark 3.2 (Approximation order). The Kraus operators in (2)–(3) are the first-order ($O(\Delta t)$) approximation of the exact GKSL propagator $\Phi_{\Delta t} = e^{\Delta t \mathcal{L}}$. The truncated channel differs from the exact channel by $O(\Delta t^2) \approx 10^{-4}$ (for $\Delta t = 0.01$). The Stinespring dilation constructed in TE2.4 is exact for this truncated channel; it agrees with the dilation of the full GKSL channel to $O(\Delta t^2)$ precision.

3.3 The JT-like PSC Model

Jackiw-Teitelboim (JT) gravity [7, 20] is a 1+1-dimensional dilaton gravity model that captures key features of near-extremal black holes and has served as a theoretical laboratory for the information paradox [12]. We use a JT-like PSC toy universe that retains the qualitative structure—a system Hilbert space $\mathcal{H}_{\text{sys}}$ representing degrees of freedom near the black hole horizon, a Hawking temperature T_H, and thermal emission and absorption Lindblad operators—while operating in a tractable finite-dimensional setting.

The system consists of $n = 3$ bosonic modes, each truncated to $d = 2$ Fock levels (vacuum $|0\rangle$ and single-excitation $|1\rangle$), giving total Hilbert space dimension $d_{\mathcal{H}} = d^n = 8$. The Hamiltonian is a sum of harmonic oscillators at frequencies $\omega_j = (j + \frac{1}{2})\pi T_H$, $j = 0, 1, 2$, with $T_H = 0.003979$ in natural units where the fundamental frequency is 1 (so that mode frequencies are $\omega_j = (j + \frac{1}{2})\pi T_H$). The lowest mode frequency is $\omega_0 = \frac{1}{2}\pi T_H \approx 0.00625$, giving $T_H/\omega_0 \approx 0.64$. The thermal occupation of the lowest mode is $\bar{n}_0 = 1/(e^{\omega_0/T_H} - 1) \approx 0.26$, placing the system in a moderate-temperature regime.

For each mode j, two Lindblad operators model the Hawking process:

$$L_{\text{emit}}^{(j)} = \sqrt{\gamma_{\text{emit},j}}\, a_j, \tag{4}$$

$$L_{\text{abs}}^{(j)} = \sqrt{\gamma_{\text{abs},j}}\, a_j^\dagger, \tag{5}$$

where a_j, $a_j^\dagger$ are the lowering and raising operators for mode j, and the rates satisfy the detailed balance condition $\gamma_{\text{emit},j}/\gamma_{\text{abs},j} = e^{-\omega_j/T_H}$, with coupling strength $\gamma_0 = 0.01$. The total number of Lindblad operators is $m = 2n = 6$, giving

$$\dim \mathcal{H}_E = 1 + m = 7. \tag{6}$$

The detailed balance condition ensures the GKSL dynamics drives the system toward the Gibbs thermal state $\rho_{\text{th}} \propto e^{-H/T_H}$, analogous to a black hole relaxing to a thermal Hawking state.

4 TE2.4 Computational Results

Experiment TE2.4 [13] implements the programme described above. All code is archived in the ugp-physics repository [18]; the canonical result file is `results/phase2_3_final/final_results.json`; the SHA-256 hash is listed in Appendix C.

4.1 Validation of CPTP Property

The GKSL channel $\Phi_{\Delta t}$ with $\Delta t = 0.01$ was verified to be CPTP by checking that, for all test states (vacuum, thermal, Fock $|1, 0, 0\rangle$), the evolved state satisfies: (i) trace preservation to error $< 10^{-10}$; (ii) all eigenvalues $\geq -10^{-10}$ (positive semidefiniteness); and (iii) Hermiticity error $< 10^{-10}$. Detailed balance was verified to 0.00% relative error.

4.2 Stinespring Construction and Fidelity

The Stinespring dilation was constructed explicitly as follows. For time step $\Delta t = 0.01$, the Kraus operators K_0 and $K_1, \ldots, K_6$ were computed using equations (2)–(3). The isometry $V : \mathcal{H}_{\text{sys}} \to \mathcal{H}_{\text{sys}} \otimes \mathcal{H}_E$ was assembled as the block-column matrix

$$V = \begin{pmatrix} K_0 \\ K_1 \\ \vdots \\ K_6 \end{pmatrix}, \tag{7}$$

a 56×8 complex matrix (8×7 block rows of width 8). The unitary dilation U was obtained by completing V to a square unitary on $\mathcal{H}_{\text{sys}} \otimes \mathcal{H}_E$ via QR-based Gram-Schmidt extension.

To verify the dilation, the evolved state was computed via both the GKSL propagator and the Stinespring construction (partial trace of $U(\rho \otimes |0\rangle\langle 0|_E)U^\dagger$), and the fidelity

$$F(\rho, \sigma) = \left(\text{Tr}\sqrt{\sqrt{\rho}\, \sigma \sqrt{\rho}} \right)^2 \tag{8}$$

was computed between the two results for three representative states: the vacuum state $|0,0,0\rangle\langle 0,0,0|$, the Gibbs thermal state at T_H, and the Fock state $|1,0,0\rangle\langle 1,0,0|$.

The results, as recorded in `final_results.json`, are:

Test State	F	Status
Vacuum $\|0,0,0\rangle$	≈ 1.0	PASS
Fock $\|1,0,0\rangle$	1.0	PASS
Thermal ρ_{th}	1.0	PASS

All three fidelities meet and exceed the threshold $F \geq 1 - 10^{-8}$ (the reported numerical values are ≥ 1.0 to double-precision floating point). The slight super-unity value for the vacuum state ($F = 1.0000000000000002$) is a rounding artefact of floating-point arithmetic and lies within double-precision machine epsilon ($\epsilon_{\text{mach}} \approx 2.2 \times 10^{-16}$).

Remark 4.1. The fidelity $F \geq 1 - 10^{-8}$ is the verification threshold used in the experiment; the actual numerical results saturate this threshold at the level of double-precision arithmetic. This reflects the exactness of the Kraus construction at the algebraic level, not merely an approximate numerical agreement.

4.3 Thermalization Entropy Curve

Starting from the vacuum state, the GKSL dynamics was run for $t \in [0, 200]$ with time step $\delta t = 0.05$. The steady-state fidelity with the Gibbs thermal state ρ_{th} is

$$F_{\text{th}} = 0.9999192951, \tag{9}$$

confirming near-perfect thermalization. The steady-state purity is 0.7142 and the von Neumann entropy is 0.4945 nats.

The entanglement entropy was tracked along the evolution. Key values:

$$\begin{aligned} S(0) &= 0 \quad \text{(vacuum)}, \\ S_{\text{peak}} &= 0.4462 \text{ nats} \quad \text{at } t = 200, \\ S_{\text{thermal}} &= 0.4592 \text{ nats}, \\ S_{\text{peak}}/S_{\text{thermal}} &= 0.9717. \end{aligned}$$

The entropy rises monotonically from zero toward the thermal value, reaching 97.2% of S_{thermal} at $t = 200$. This is qualitatively consistent with the early-time portion of a Page curve [11]: entropy rises as Hawking radiation entangles with the system. In this toy model the entropy does not decrease back to zero because the finite-dimensional GKSL dynamics relaxes to a mixed thermal fixed point, not a pure state—the Page curve turnover is expected only in models where the black hole completely evaporates.

Remark 4.2 (Fidelity Distinction). We distinguish two fidelity quantities throughout. The Stinespring fidelity F (Table above) measures agreement between the GKSL channel and its unitary dilation; this is $\geq 1 - 10^{-8}$. The thermalization fidelity $F_{\text{th}} = 0.9999192951$ (equation (9)) measures how closely the steady state matches the ideal Gibbs state. These are different quantities addressing different aspects of the model.

4.4 What TE2.4 Establishes

To summarise precisely what the experiment proves:

1. *Mathematical fact (known):* Every CPTP channel admits a Stinespring dilation [19]. The GKSL evaporation channel $\Phi_{\Delta t}$ is CPTP (verified numerically), so a unitary dilation U exists.

2. *Explicit construction (new):* An explicit Kraus-based Stinespring construction is given for the JT-like PSC model with $\dim \mathcal{H}_E = 7$, verified to fidelity $F \geq 1 - 10^{-8}$ on three representative states.

3. *Physical interpretation (new, within MFRR):* The operator PT^{-1} is identified as the physical realisation of the recovery channel corresponding to this unitary dilation, within the PSC framework.

4. *Thermalization and entropy curve (new):* The JT-like PSC model thermalises to within 0.008% of the Gibbs state. The open Markovian GKSL model, which models early-time Hawking evaporation into an infinite bath, exhibits a monotonically rising entropy consistent with the early-time Page behaviour; entropy does not turn over because the radiation register is unbounded (see § 6).

5. *Operational* PT^{-1} *circuit—open-system model (new):* The operator PT^{-1} is constructed explicitly as the 56-dimensional quantum circuit $U^\dagger$ (the Stinespring adjoint). Applying $U^\dagger$ to the system+environment pair recovers the initial vacuum state to fidelity $F = 0.9967$; the recovered

state has purity 0.993 (SHA-256 `882cd67a...`, `results/pt_inverse_circuit/`).

6. *Full Page curve and exact* PT^{-1} *recovery—unitary closed-system model, pure initial BH (new):* In the physically correct closed model—BH as n qubits in pure state $|+\rangle^{\otimes n}$, radiation as n fresh qubits in $|0\rangle^{\otimes n}$, evaporation via unitary partial-SWAP operators U_{evap}—we obtain the following for $n = 4$ modes (SHA-256 `7f132bb1...`, `results/page_curve_unitary/`):
   - S_{BH} rises to 0.2458 nats at the Page time, then falls to 0.000000 nats: the *full Page curve with turnover*.
   - $S_{\mathrm{BH}} = S_{\mathrm{rad}}$ throughout, with total purity = 1.000000, confirming the evolution is exactly unitary.
   - Applying $\mathrm{PT}^{-1} \equiv U_{\mathrm{evap}}^{\dagger}$ recovers the initial state to fidelity $F = 1.000000$ (exact by unitarity).

   The finite radiation register is essential: once $\dim \mathcal{H}_{\mathrm{rad\,filled}} > \dim \mathcal{H}_{\mathrm{BH}}$, complementarity of the pure total state forces S_{BH} to decrease. This is the standard Page [11] argument realised explicitly within the MFRR framework.

7. *Purified thermal BH—Page curve and exact* PT^{-1} *recovery (new):* To make the closed model directly comparable to the initial thermal state studied in the GKSL model, we introduce a reference system $\mathcal{H}_R \cong \mathcal{H}_{\mathrm{BH}}$ that purifies the thermal BH density matrix. The total initial state is $|\Phi^+\rangle_{\mathrm{BH},R} \otimes |0\rangle_{\mathrm{rad}}^{\otimes n}$, where $|\Phi^+\rangle$ is the maximally entangled state (i.e. the thermal BH is purified by its interior degrees of freedom R); R remains untouched during evaporation. This is precisely the setup underlying the island formula. For $n = 3$ modes (SHA-256 `e9ac80b0...`, `results/page_curve_thermal/`; confirmed independently for $n = 4$, SHA-256 `3cbd71f2...`):
   - S_{BH} starts at $\ln 8 = 2.079$ nats (thermal entropy), decreases monotonically as radiation absorbs BH entropy, and reaches 0 after full evaporation: the *Page curve for a thermal BH*.
   - The Page crossing occurs when $S_{\mathrm{rad}} = S_{\mathrm{BH}}$ after $n/2$ modes have been emitted, in exact agreement with Page [11]. For even n this is an integer mode count (e.g. $n = 4$: crossing at exactly 2 modes); for odd n (here $n = 3$), $n/2 = 1.5$ is non-integer and the crossing occurs continuously between the $\lfloor n/2 \rfloor$-th and $\lceil n/2 \rceil$-th discrete mode emissions (confirmed numerically at $t = 30$ of 60 simulation steps, i.e. 1.5 modes).
   - $S_R = \ln 8$ throughout; S_{rad} rises to $\ln 8$ at the end, encoding all the original BH thermal entropy. Total purity = 1.000000.
   - $\mathrm{PT}^{-1} \equiv U_{\mathrm{evap}}^{\dagger}$ restores the BH–R maximally entangled state to fidelity $F = 1.000000$, recovering $S_{\mathrm{BH}} = \ln 8$: the thermal entropy is exactly returned.

Items 1–2 are established within the toy model. Item 3 is a physical interpretation within MFRR. Items 4–5 describe the Markovian (open) regime; item 6 provides the full closed-system result for a pure BH; item 7 extends it to the physically relevant thermal BH in the purified setup underlying the island formula. None constitutes a proof that information is preserved in real astrophysical black holes.

TE2.4 does not: (1) prove information preservation in real astrophysical black holes; (2) provide a first-principles derivation of PT^{-1} independent of the Stinespring construction; (3) engage with the firewall paradox at the level of local field theory. Item 5 uses the first-order Kraus approximation ($O(\Delta t^2)$ precision); items 6–7 use exact unitary circuits on finite qubit models.

Remark 4.3 (Open, closed-pure, and closed-thermal models). The open GKSL model (items 1–5) is a Markovian approximation to early-time Hawking evaporation; its infinite bath prevents Page-curve turnover. The closed pure-state model (item 6) establishes the full Page curve for a BH with initially zero entropy. The closed thermal model (item 7) is the physically representative case: the BH starts fully thermalized, the initial entropy is carried by entanglement with a reference R (representing the BH interior), and full evaporation transfers all entropy to radiation while PT^{-1} restores the original BH–R entanglement exactly. All three models are consistent with the Stinespring/MFRR framework and address complementary physical regimes.

5 Relationship to Existing Approaches

5.1 Island Formula and Quantum Extremal Surfaces

The island formula [1,12] derives the Page curve in JT gravity coupled to a CFT bath by identifying "island" regions whose entropy is included in the fine-grained entropy of Hawking radiation via the quantum extremal surface prescription. This reproduces the Page curve turnover without requiring direct knowledge of the microscopic recovery channel.

The MFRR approach differs in emphasis. The island calculation works in the semiclassical path integral and requires a replica trick; the MFRR approach works in the operator-algebraic language of CPTP maps and makes the recovery channel explicit via PT^{-1}. The two frameworks are complementary contributions to the information paradox: the island formula delivers the thermodynamic entropy curve establishing that information is not lost; the MFRR approach provides the explicit operator-algebraic recovery mechanism at the Hilbert-space level. A complete resolution of the information paradox would benefit from both: the entropy curve (from island/QES calculations) and the recovery operator (from MFRR/Stinespring). The present work contributes the latter, for a toy model.

A notable limitation of our current model, by comparison with the island formula, is that the JT-like PSC model is fully finite-dimensional and does not contain the gravitational path integral structure that gives rise to replica wormholes. Extension to that regime is an open problem.

5.2 ER=EPR

Maldacena and Susskind [9] proposed that maximally entangled pairs of particles are connected by Einstein-Rosen bridges, suggesting that entanglement and wormholes are dual descriptions of the same phenomenon. The MFRR approach does not contradict ER=EPR; both posit that information is geometrically or structurally encoded in the environment degrees of freedom rather than destroyed. The PT^{-1} mechanism can be seen as providing a dynamical account of the recovery process whose existence ER=EPR motivates.

5.3 Firewall Paradox

The AMPS firewall argument [2] shows that maintaining unitarity, the equivalence principle, and the no-drama condition for infalling observers is inconsistent within standard semiclassical assumptions. The MFRR framework does not resolve the firewall paradox: we do not provide a model for the experience of an infalling observer, nor do we address the equivalence principle at the level of local field theory. The present work operates entirely at the level of the system Hilbert space and its environment dilation; it does not engage with the local vs. global description tension that underlies the firewall.

5.4 Page Curve

The open GKSL model (items 1–4, § 4) exhibits an entropy curve rising from zero toward the thermal value, consistent with early-time Hawking evaporation. The turnover is absent because the infinite Markovian bath has a fixed-point attractor at ρ_{th}; it is not a counter-evidence to information preservation but a feature of open-system dynamics.

The full Page curve — both the rise and the fall to zero — is exhibited in two complementary closed unitary models. Item 6 (pure initial BH, $n = 4$ qubits) demonstrates entropy turnover and exact PT^{-1} recovery ($F = 1.000000$). Item 7 (purified thermal initial BH, $n = 3$ and $n = 4$ qubits, the island formula setup) demonstrates the full thermal Page curve: S_{BH} starts at $\log d_{\mathrm{BH}}$, transfers monotonically to radiation with the Page crossing at $n/2$ emitted modes (exactly as predicted by Page [11]), and reaches zero at full evaporation. $\mathrm{PT}^{-1} = U_{\mathrm{evap}}^{\dagger}$ then restores the BH–interior entanglement exactly ($F = 1.000000$).

The recovery is exact because the closed model is by construction unitary: $U_{\mathrm{evap}}^{\dagger} U_{\mathrm{evap}} = I$. This is not a proof that information is preserved in real astrophysical black holes, but it demonstrates that within a finite-dimensional unitary framework consistent with MFRR/PSC, the complete Page curve and exact information recovery are achieved in the physically representative thermal BH setup.

6 Scope, Limitations, and Open Fronts

6.1 Toy Model Limitations

The JT-like PSC model is a finite-dimensional, Markovian, non-relativistic toy model. It captures the qualitative structure of Hawking radiation (thermal emission and absorption, detailed balance, CPTP dynamics) but omits:

- The gravitational path integral and its non-perturbative saddles (replica wormholes, gravitational instantons).
- The continuous spectrum of a genuine quantum field theory.
- Back-reaction of radiation on the black hole geometry.
- Relativistic and causal structure of the horizon.

Claims made in this paper are restricted to the toy model and its mathematical properties.

6.2 The PT^{-1} Physical Realization

The PT^{-1} operator is defined at the level of the MFRR framework as a structural constraint on PSC-consistent histories. We have not provided a Hilbert-space operator realisation of PT^{-1} independent of the Stinespring construction. Establishing whether PT^{-1} can be identified with a specific quantum circuit or gravitational operation in a more realistic model is an open problem.

6.3 Extension to 3+1 Gravity

The most important open front is the extension to 3+1-dimensional quantum gravity. In a realistic black hole, the system Hilbert space is infinite-dimensional, the dynamics is not Markovian at all timescales, and the notion of a preferred Hilbert space decomposition $\mathcal{H}_{\mathrm{sys}} \otimes \mathcal{H}_E$ is itself ambiguous (the factorisation problem). Addressing these issues requires tools beyond those employed here.

6.4 Non-Markovian Effects and Long-Time Unitarity

The Stinespring dilation is constructed here for a single time step $\Delta t = 0.01$. Over long timescales, GKSL dynamics with a fixed Lindbladian is Markovian by assumption; genuine black hole evaporation may exhibit non-Markovian memory effects that are not captured by the GKSL model. The relationship between the step-wise Stinespring construction and a global unitary for the full evaporation history is an open technical question.

6.5 Falsifiability

The MFRR framework is falsifiable in the following senses. (i) If the PSC condition were shown to be inconsistent with Standard Model physics at low energies, the framework would be refuted (this has been examined in other papers in the programme [17]). (ii) If a no-go theorem were established showing that no Hilbert-space operator can implement PT^{-1} in a relativistic setting consistent with causality, the interpretive claim would be undermined. (iii) The specific numerical predictions of the toy model (thermalization fidelity, entropy curve, Stinespring dimension) are falsifiable within the model and have been verified in TE2.4.

The abstract PSC record-consistency theorem provides machine-checked formal backing for this framework: NEMS Paper 38 [15] proves `BlackHoles.record_consistency_abstract` in nems-lean (Zenodo 10.5281/zenodo.19429791, zero `sorry`) — if two worlds disagree on an observation, the semantics is non-categorical and a selector is required, ruling out genuine information destruction in a PSC universe. The no-hypercomputing-from-black-holes theorem (NEMS 35+38 [15, 16], `BlackHoles.no_hypercomputing_from_bh`, zero `sorry`) further establishes that no total-effective black-hole decoder can decide a diagonal-capable predicate on the outcome space.

7 Conclusion

We have examined the black hole information paradox through the lens of the MFRR framework, proposing that *reflexive unitarity*—the structural requirement imposed by the PSC condition together with the canonical operator PT^{-1}—provides a physical interpretation of the unitary dilation guaranteed by the Stinespring theorem for any CPTP Hawking evaporation channel.

The key mathematical fact is established: GKSL evaporation channels are CPTP and therefore admit

Stinespring dilations. The novel contributions are:

1. The MFRR interpretation of PT^{-1} as the explicit recovery operator identified with the Stinespring adjoint $U^\dagger$.

2. The explicit Stinespring construction for a JT-like PSC universe with $\dim \mathcal{H}_E = 7$, verified to fidelity $F \geq 1 - 10^{-8}$ and thermalization fidelity $F_{\mathrm{th}} = 0.9999193$ (TE2.4).

3. An explicit PT^{-1} quantum circuit recovering the vacuum from a partially-thermalized state to $F = 0.9967$ in the open model.

4. The full Page curve with turnover and exact PT^{-1} recovery ($F = 1.000000$) in a closed finite-qubit model (pure initial BH).

5. The complete Page curve for the *thermal* initial BH in the island-formula setup (BH purified by reference R representing the BH interior): entropy transfers monotonically from BH to radiation with the Page crossing at $n/2$ modes (exactly as Page [11] predicted), and $\mathrm{PT}^{-1} = U_{\mathrm{evap}}^\dagger$ restores the BH–interior entanglement to $F = 1.000000$.

We have been explicit about what this work does not establish: it does not prove that information is preserved in real astrophysical black holes, does not resolve the firewall paradox, and does not provide a first-principles derivation of PT^{-1} beyond identification with the Stinespring adjoint. These remain open problems requiring extension to 3+1 quantum gravity and non-Markovian dynamics.

The MFRR programme provides a structural framework in which the question "which unitary implements information recovery?" has a principled and constructive answer: it is $U_{\mathrm{evap}}^\dagger$, the adjoint of the evaporation unitary, identified physically as PT^{-1}. Within the toy model this answer is exact and reproducible. Translating this principle into a concrete prediction for astrophysical black holes is the primary open front of this programme.

A JT-like PSC Model: Technical Parameters

The model parameters, as used in TE2.4 [13], are:

Parameter	Symbol	Value
Number of modes	n	3
Fock levels per mode	d	2
System dimension	$d_{\mathcal{H}}$	$8 = 2^3$
Hawking temperature	T_H	0.003979
Coupling strength	γ_0	0.01
Mode frequencies	ω_j	$(j + \frac{1}{2})\pi T_H$
Lindblad operators	m	6 (2 per mode)
Env. dimension	$\dim \mathcal{H}_E$	$7 = 1 + m$
Time step	Δt	0.01
Evolution duration	$t_{\max}$	200

The Lindblad operators for mode j are

$$L_{\mathrm{emit}}^{(j)} = \sqrt{\gamma_{\mathrm{emit},j}}\, I^{\otimes j} \otimes a \otimes I^{\otimes(n-j-1)},$$
$$L_{\mathrm{abs}}^{(j)} = \sqrt{\gamma_{\mathrm{abs},j}}\, I^{\otimes j} \otimes a^\dagger \otimes I^{\otimes(n-j-1)},$$

where $a = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ in the two-level basis $\{|0\rangle, |1\rangle\}$, and the rates satisfy $\gamma_{\mathrm{emit},j}/\gamma_{\mathrm{abs},j} = e^{-\omega_j/T_H}$. All computations use double-precision floating-point arithmetic (IEEE 754, $\epsilon_{\mathrm{mach}} \approx 2.2 \times 10^{-16}$).

B Stinespring Construction: Algorithmic Details

Given the GKSL master equation with Hamiltonian H and Lindblad operators $\{L_k\}_{k=1}^m$ at coupling strengths $\{\gamma_k\}$, the Stinespring construction for time step Δt proceeds as follows:

1. Compute $K_0 = I - (iH + \frac{1}{2}\sum_k \gamma_k L_k^\dagger L_k)\Delta t$.

2. Compute $K_k = \sqrt{\gamma_k \Delta t}\, L_k$ for $k = 1, \ldots, m$.

3. Verify completeness: $\sum_{k=0}^m K_k^\dagger K_k \approx I$ (error of order Δt^2; for $\Delta t = 0.01$ this is $\sim 10^{-4}$, consistent with first-order Kraus approximation).

4. Assemble the isometry V as in equation (7): a $(d_{\mathcal{H}}(m+1)) \times d_{\mathcal{H}}$ matrix where the k-th block of $d_{\mathcal{H}}$ rows is K_k.

5. Extend V to a unitary U on the full space $\mathcal{H}_{\mathrm{sys}} \otimes \mathcal{H}_E$ (dimension $d_{\mathcal{H}} \times (m+1) = 56$) via QR decomposition and Gram-Schmidt completion of the orthogonal complement.

6. Verify: for each test state ρ, compute $\rho' = \mathrm{Tr}_E[U(\rho \otimes |0\rangle\langle 0|)U^\dagger]$ and compare with $\Phi_{\Delta t}(\rho) = \sum_k K_k \rho K_k^\dagger$ by computing the fidelity $F(\rho', \Phi_{\Delta t}(\rho))$.

The first-order Kraus approximation (steps 1–2) is valid for small Δt; for the exact propagator $\Phi_t = e^{t\mathcal{L}}$ the Kraus operators are obtained from the matrix exponential of the superoperator representation of $\mathcal{L}$.

C Reproducibility

All computational results reported in this paper are produced by the TE2.4 experiment in the ugp-physics repository [18]. The canonical results are:

Result	Script	SHA-256
GKSL (items 1–4)	`te2_4_final_production.py`	`bf2b079c...`
PT^{-1} circ. (5)	`te2_4_pt_inverse_circuit.py`	`882cd67a...`
Page curve pure (6)	`te2_4_page_curve_unitary.py`	`7f132bb1...`
Page curve therm. (7)	`te2_4_page_curve_thermal.py`	`e9ac80b0...`
($n = 4$ ext.)		`3cbd71f2...`

All scripts reside in `MFRR/TE_2_Advanced_Explorations/TE_2_4_BH_Unitarity/src/`; results in the corresponding `results/` subdirectories. Language: Python 3, NumPy, SciPy. Full SHA-256 values and run parameters are recorded in `papers/16_bh_unitarity/PROVENANCE.md`.

D Lean Formalization of PSC Constraints on Black Hole Information

Two results in the nems-lean library (Zenodo 10.5281/zenodo.19429710, GitHub https://github.com/novaspivack/nems-lean) provide machine-checked formal backing for the claims of this paper. Both are proved with zero `sorry` and zero custom axioms in these proof chains; the build command is `lake update && lake build` from the nems-lean root.

`BlackHoles.record_consistency_abstract` (NEMS Paper 38, Zenodo 10.5281/zenodo.19429791). *Statement:* Given a semantic structure where world w satisfies observation o but world w' does not (an "information-erasing appearance"), the observation semantics is non-categorical: at least two world-types are distinguishable, and a selector is required. Under PSC (which forbids external selectors), this rules out genuine information destruction in a PSC universe. *Key proof step:* The proof formalises that a categorical semantics would make w and w' observationally equivalent, contradicting the erasing appearance. The conclusion that an internal selector must exist then follows from the NEMS trichotomy.

`BlackHoles.no_hypercomputing_from_bh` (NEMS Papers 35+38, Zenodo 10.5281/zenodo.19429785). *Statement:* Under the SelectorStrength barrier premises (extensional nontrivial predicate, anti-decider closure, fixed-point hypothesis), no total-effective black-hole decoder can decide a diagonal-capable predicate on the BH outcome space. *Key proof step:* Instantiates the general `SelectorStrength.no_total_decider_at_strength_nontrivial` barrier theorem (proved by diagonalisation) to the black-hole decoding predicate class.

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